Chung-Hsuan "Andy" Tung

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Education

DUKE UNIVERSITY, Department of Electrical and Computer Engineering — Durham, NC
Ph.D. Candidate — 2021–2026 (expected)
Research Interests: High-Performance Computing, Hardware–Software Co-design, Hardware Reliability, Wireless Signal Processing
Overall GPA: 3.97/4

NATIONAL CHENG KUNG UNIVERSITY (NCKU), Department of Electrical Engineering — Tainan, Taiwan
Bachelor of Science — 2016–2020
Grade: 93.46/100 (Overall GPA: 4.22/4.3, Ranked 1/155)

Internships

NVIDIA CORPORATION — Santa Clara, CA
Resilience and Safety Architecture Intern — May–Aug. 2025
- Analyzed gate-level fault simulation results to characterize silent data corruption (SDC) patterns in GPUs
- Proposed statistical abstractions of SDC behaviors to inform higher-level fault injection and modeling

Design for Test (DFT) Intern (remote) — May–Oct. 2022
- Scripted to convert In-System Test specifications to RTL-level design for ISO 26262 safety requirements
- Contributed to interpreting reliability, availability, and serviceability (RAS) related logs

XILINX INC. , a subsidiary of ADVANCED MICRO DEVICES INC. (AMD), AI Engine Driver Team — San Jose, CA
Machine Learning Intern — May–Aug. 2024
- Developed bare-metal control and kernels for the AI Engine in the AMD Versal adaptive SoCs
- Automated the Vitis build flow for the AI Engine bare-metal applications

Selected Publications

- **C.-H. Tung**, Y. Huang, N. Saxena, P. Shirvani, S. Hukerikar, T. Jain, A. Tyagi, and S. Gongalore, "The Anatomy of Silent Data Corruption: GPU Error Pattern Study and Modeling Guidance," *IEEE/IFIP International Conference on Dependable Systems and Networks — Industry Track (DSN–S), 2026* (with Nvidia) **[arXiv]**
- J. Zheng*, **C.-H. Tung***, Y. Yao, N. Mehrotra, S. Mattu, Z. Qi, D. Zhuo, R. Calderbank, and T. Chen, "Real-Time and Scalable Zak-OTFS Receiver Processing on GPUs," *(under review) 2026* (*co-first author) **[arXiv] [GitHub]**
- **C.-H. Tung**, Z. Qi, and T. Chen, "RISE: Real-time Image Processing for Spectral Energy Detection and Localization," *IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN), 2026* **[Paper] [GitHub]**
- Z. Qi, Y. Yao, Y. Li, **C.-H. Tung**, J. Zheng, D. Zhuo, and T. Chen, "DecodeX: Exploring and Benchmarking of LDPC Decoding across CPU, GPU, and ASIC Platforms," *ACM International Workshop on Mobile Computing Systems and Applications (HotMobile), 2026* **[Paper]**
- Z. Qi, **C.-H. Tung**, Z. Gao, and T. Chen, "Nexus: Efficient and Scalable Multi-Cell mmWave Baseband Processing with Heterogeneous Compute," *ACM International Conference on Mobile Computing and Networking (MobiCom), 2026* **[arXiv] [GitHub]**
- Z. Qi*, **C.-H. Tung***, A. Kalia, and T. Chen, "Savannah: Efficient mmWave Baseband Processing with Minimal and Heterogeneous Resources," *ACM International Conference on Mobile Computing and Networking (MobiCom), 2024* (*co-first author) **[Paper] [GitHub]**
- **C.-H. Tung**, B. K. Joardar, P. P. Pande, J. R. Doppa, H. Li, and K. Chakrabarty, "Dynamic Task Remapping for Reliable CNN Training on ReRAM Crossbars," *Design, Automation & Test in Europe Conference & Exhibition (DATE), 2023* **[Paper]**

Research Experience

DUKE UNIVERSITY, FuNCtions Laboratory — Durham, NC
Graduate Researcher & SRC Research Scholar (Advisor & PI: Tingjun Chen, Ph.D.) — Jan. 2023–Now

Neural receiver integration into real-time 5G vRAN testbed
- Implemented a RNN-based MIMO-OFDM receiver within a multi-threaded C++ baseband framework
- Identified matrix inversion/multiplication as dominant bottlenecks (65–80% of end-to-end latency)
- Demonstrated architectural limits preventing real-time scalability under current design

Real-time Spectrum Sensing with Energy Detection and Localization using Image Processing Techniques
- Developed a pipeline using adaptive thresholding and morphological operation to detect signal occupancy
- Implemented multi-threaded architecture to achieve real-time 100 MHz bandwidth sensing
- Explored YOLO object detection and U-Net semantic segmentation with self-attention as ML baselines

Real-time Baseband Processing for mmWave Signals (5G FR2 $\mu 3$) in vRAN
- Optimized matrix operations (performance bottlenecks in MIMO DSP) with SIMD intrinsics (AVX-512)
- Implemented DSP in multi-threaded C++ to support 487 Mbps data rate with one CPU core and an eASIC

DUKE UNIVERSITY, Emerging Computing Technologies Laboratory — Durham, NC

Graduate Researcher (Advisor: Krishnendu Chakrabarty, Ph.D.) — Aug. 2021–Dec. 2022

Reliability of ReRAM Crossbar-based CNN Accelerator

- Explored impacts of stuck-at faults (SAFs) in ReRAM-based accelerators on CNN training accuracy
- Developed a built-in self-test (BIST) procedure as a reliable fault detection method
- Proposed dynamic task remapping to mitigate the impact of SAFs by inherent fault tolerance of CNN models

NATIONAL CHENG KUNG UNIVERSITY, Computer Architecture and System Laboratory (CASLab) — Tainan, Taiwan

Undergraduate Researcher (Advisor: Chung-Ho Chen, Ph.D.) — Oct. 2018–Nov. 2019

The Evaluation of Implementing AI Hardware Accelerator on Embedded System

- Ported GPU code for Micro Darknet for Inference (MDFI) framework developed by CASLab to reduce the execution time by 95% with YOLOv3
- Pruned LeNet, tested it on the ARC IoT IoTDK with MDFI CPU version, and maintained 73% accuracy in 128 kB memory usage

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Skills

Languages	C/C++, Verilog, SystemC, Python, Bash Script, MATLAB, Java, LaTeX, Perl
Software	Vim, GitHub, CMake, Make, PyTorch, PytorX, NeuroSim, BookSim, OpenMP, Lex, Yacc, LabVIEW, SimpleScalar
EDA Tools	Cadence NC-Verilog, JasperGold Superlint, Incisive Comprehensive Coverage (ICC); Synopsys Design Vision, Design Compiler, Laker; Mentor Graphics Pyxis, ModelSim

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Relevant Coursework

Graduate	Advanced Computer Architecture I, Performance Optimization and Parallelism, Computer Engineering ML and Deep Neural Nets, CMOS VLSI Design Methodologies, VLSI System Testing, Digital Integrated Circuits, Advanced Computer Network
Undergrad	Computer Organization, VLSI System Design, Compiler Construction, AI-on-chip for Machine Learning and Inference, Advanced Topics in Electronic System Level Design, Operating System, Computer Algorithm, Internet Architecture, Logic System, Electrical Circuits, Electronics

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Awards and Honors

Student Leadership Award from Center for Ubiquitous Connectivity (CUbiC), SRC — Jun. 2024

- In recognition of outstanding research achievements and accomplishments during the 2024 CUbiC Annual Review

Honorary Member of The Phi Tau Phi Scholastic Honor Society of the Republic of China — Since Jun. 2020

- For ranking among the top 1% (1/155) undergraduate graduands in academic performance at Dept. of EE, NCKU

Outstanding Student For the Academic Achievement — 2016–2020

- For ranking among the top 10% in academic performance at Dept. of EE, NCKU

Outstanding Winner at 2018 Synopsys ARC Design Contest — Jan.–Jul. 2018

- Was the leader of a four-member team that ranked among the top 10 of about 100 international teams
- Constructed a vending machine with cloud management and smart selling systems and was responsible for IoT hardware control

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Professional Service

Session Chair, IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), *Industry Track* — Jun. 2026

Reviewer, IEEE Wireless Communication Magazine (WCM), IEEE Journal on Selected Areas in Communications (JSAC) — 2026

Leadership

President, *Duke Taiwanese Student Association (DTSA)*, Duke University, Durham, NC — 2022–2023

Undergraduate Student Representative, *EECS College Faculty Meetings*, NCKU, Tainan, Taiwan — Jul. 2019–Jun. 2020

Selected Course Projects

Full-custom Digital Signal Processing Unit, *CMOS Design Methodologies* — Sep.–Dec. 2021

- Designing a 3-tap 8-bit finite impulse response filter with booth multipliers and carry-lookahead adders in a full-custom approach

***μ*Go Compiler Implementation,** *Compiler Construction* — Apr.–Jun. 2020

- Designed a compiler to parse a Go-like language and generate Java machine code with Lex and Yacc

Verilog Implementation of a Synthesizable Pipelined MIPS II Processor, *VLSI System Design* — Dec. 2019–Jan. 2020

- Designed a 5-stage pipelined synthesizable MIPS II processor in RTL coding style, along with two team members
- Verified design rule eligibility by SuperLint and testbench completeness by ICC
- Synthesized and simulated at a 25 MHz clock rate with an area of 0.2 mm^2 and power of 3 milliwatts by Design Vision